

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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"Fitness and Diet Tracker with Personalized Recommendations"

Agile Sprint Planning & User Story Workshop Report

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ABSTRACT

The fitness and wellness industry has experienced significant growth due to increasing awareness of healthy lifestyles and preventive healthcare. People seek digital solutions to monitor physical activities, calorie intake, and overall well-being. Fitness centers, nutritionists, and healthcare providers require integrated platforms to offer personalized health guidance. Wearable devices and mobile applications generate valuable health data that can be analyzed for personalized recommendations. AI-powered analytics help users achieve fitness goals through customized exercise and diet plans. However, many existing applications lack comprehensive personalization and progress tracking. A smart fitness and diet tracker can bridge this gap by providing data-driven recommendations and continuous health monitoring.

This report presents an Agile Sprint Planning and User Story Workshop for the Fitness and Diet Tracker platform. The workshop defines the project vision and stakeholders, decomposes requirements into Epics and User Stories, specifies measurable acceptance criteria in Given-When-Then format, prioritizes the product backlog, plans Sprint 1, estimates effort using Planning Poker with the Fibonacci sequence, and defines the Sprint Goal and expected deliverables, demonstrating how Agile practices deliver incremental, testable, business-aligned value for a complex AI/IoT-driven health and wellness platform.

Keywords: Agile, Sprint Planning, User Stories, Fitness Tracker, Diet Recommendation, AI Analytics, Wearable Integration, Health Monitoring, Scrum.

1. INTRODUCTION

The global health and wellness sector is undergoing a digital transformation, driven by an increasing awareness of preventive healthcare and active lifestyle management. Modern consumers increasingly rely on digital applications and wearable devices to capture daily biometric data, physical activity levels, and dietary habits. However, integrating diverse streams of personal health data into cohesive, actionable guidance remains a key challenge in mobile health platform engineering.

Problem Statement: Design and develop a Fitness and Diet Tracker that enables users to monitor daily physical activities, calorie intake, and nutritional habits. The platform should collect user health data, including age, weight, BMI, and fitness goals, to generate personalized recommendations. It should provide customized workout schedules and diet plans based on individual preferences and health conditions. The system must track progress using interactive dashboards and visual reports. Integration with wearable devices should allow automatic synchronization of health metrics. The application should send reminders for workouts, hydration, and meals while ensuring data privacy and security. The solution should offer an intuitive interface to encourage consistent healthy lifestyle management.

Purpose: This report translates the Fitness and Diet Tracker problem statement into an actionable Agile plan covering stakeholder analysis, Epics/User Stories, acceptance criteria, backlog prioritization, sprint planning, and estimation ready for a Scrum team to begin development.

PROJECT VISION & STAKEHOLDERS

Vision: "To build an AI-powered, IoT-enabled Fitness and Diet Tracking Platform that empowers individuals to achieve optimal health and wellness through real-time activity tracking, automated meal and workout personalization, intelligent wearable synchronization, and continuous data-driven health monitoring."

Objectives: Develop a personalized fitness and diet management platform; track daily physical activities and nutritional intake; generate AI-based workout and diet recommendations; integrate wearable device data for health monitoring; visualize user progress through reports and dashboards; send reminders for exercise, meals, and hydration; ensure secure storage of personal health information; promote healthy lifestyle habits through continuous monitoring.

Expected Outcomes: A fully functional fitness and diet tracking application; personalized workout and meal recommendations; real-time health and activity monitoring; improved user engagement through reminders and notifications; accurate visualization of fitness progress; secure management of health-related data; better adherence to fitness and nutrition goals; enhanced overall health and wellness of users.

Stakeholder Analysis:

Stakeholder	Interest
End User / Individual Practitioner	Primary user; tracks daily calories, logs workouts, syncs wearables, and views personalized recommendations.
Nutritionist / Dietitian	Reviews client nutritional logs, customizes meal plans, and sets macro targets.
Fitness Trainer / Coach	Monitors client exercise compliance, creates tailored workout regimens, and adjusts dynamic goals.
Healthcare Provider / Physician	Accesses aggregated biometric reports (with user consent) to evaluate lifestyle impact on health conditions.
Wearable Device Vendors / Telematics	Provides IoT hardware APIs (heart rate, step counts, sleep cycles) for background data sync.
Data Science / AI Team	Builds, trains, and evaluates recommendation models for personalized diet and exercise plans.
System Administrator / Security Officer	Manages user authentication, data encryption, role-based access, and HIPAA/GDPR privacy compliance.
Product Owner	Defines business vision, prioritizes the product backlog, and accepts completed user stories.

Scrum Master	Facilitates Agile ceremonies, removes technical blockers, and enforces Scrum practices.
Development Team	Engineers front-end user interfaces, back-end microservices, and AI recommendation pipelines.

EPICS AND USER STORIES

Epic	Focus Area
E1	User Profile Management & Biometric Health Onboarding
E2	Daily Activity & Nutritional Intake Tracking
E3	AI-Powered Diet & Workout Recommendation Engine
E4	Wearable Device & IoT Data Synchronization
E5	Visual Analytics, Dashboards & Progress Reporting
E6	Smart Notifications, Reminders & Security Management

E1 - User Profile Management & Biometric Health Onboarding

- > **US1.1:** As a new user, I want to create a profile entering age, weight, height, and medical conditions, so that the system calculates my base BMI and daily caloric baseline.
- > **US1.2:** As a user, I want to set specific fitness goals (e.g., weight loss, muscle gain, maintenance), so that the application tailors targets to my intent.
- > **US1.3:** As a user, I want to specify dietary preferences and restrictions (e.g., vegan, keto, allergies), so that generated meal plans remain safe and suitable.

E2 - Daily Activity & Nutritional Intake Tracking

- > **US2.1:** As a user, I want to log my daily food intake by searching a food database or scanning barcodes, so that I can track calories and macronutrients.
- > **US2.2:** As a user, I want to manually log physical activities and strength workouts, so that my daily active energy expenditure is accurately calculated.
- > **US2.3:** As a user, I want to record daily water intake, so that I can maintain hydration goals.

E3 - AI-Powered Diet & Workout Recommendation Engine

- > **US3.1:** As a user, I want AI-generated personalized meal plans based on my biometric profile and goals, so that I meet my nutritional targets effortlessly.
- > **US3.2:** As a user, I want adaptive daily workout recommendations based on my current energy levels and activity history, so that I can train efficiently without overtraining.
- > **US3.3:** As a user, I want real-time dynamic macro adjustments when I over- or under-eat, so that my plan adapts automatically to keep me on track.

E4 - Wearable Device & IoT Data Synchronization

- > **US4.1:** As a user, I want to pair my smart wearable device (e.g., smartwatch, fitness band), so that my

heart rate, steps, and active calories sync automatically.

> **US4.2:** As a user, I want background sync of sleep quality and resting heart rate data, so that the system assesses my daily recovery status.

E5 - Visual Analytics, Dashboards & Progress Reporting

> **US5.1:** As a user, I want an interactive main dashboard displaying my real-time calorie balance, steps, and workout completion, so that I get an immediate view of daily progress.

> **US5.2:** As a user, I want weekly and monthly visual reports showing weight trends, macro ratios, and workout consistency, so that I stay motivated and monitor long-term trends.

> **US5.3:** As a user, I want to export progress reports as PDF documents, so that I can share them with my nutritionist or personal trainer.

E6 - Smart Notifications, Reminders & Security Management

> **US6.1:** As a user, I want automated push notifications for workout sessions, scheduled meals, and hydration intervals, so that I maintain continuous lifestyle consistency.

> **US6.2:** As a user, I want secure multi-factor authentication and data encryption, so that my personal biometric and health data remain private and secure.

2. ACCEPTANCE CRITERIA

US1.1: Given a newly registered user on the profile onboarding screen, When they input valid age, weight, height, and gender data, Then the system automatically calculates and displays their current BMI, basal metabolic rate (BMR), and daily caloric baseline within 1 second.

US2.1: Given a logged-in user viewing the food logger, When they search for a food item or scan a barcode, Then the system retrieves the complete nutritional profile (calories, protein, carbs, fats) from the database and updates the daily macro summary upon confirmation.

US3.1: Given a user with configured dietary preferences and fitness goals, When an AI meal plan recommendation is requested, Then the system generates a balanced 7-day meal plan matching target calories within $\pm 5\%$ and excluding all flagged allergens.

US4.1: Given an active wearable device connection, When new step count or heart rate data is recorded by the wearable sensor, Then the application automatically synchronizes the data in the background every 15 minutes without requiring user intervention.

US5.1: Given a user opening the application home screen, When biometric and activity data exist for the day, Then the interactive dashboard displays total consumed vs. burned calories, step goal progress percentage, and remaining water quota simultaneously.

US6.1: Given a user who has enabled hydration and meal alerts, When the scheduled reminder time is reached, Then the application delivers a local push notification to the user's mobile device within 30 seconds of the scheduled trigger.

US6.2: Given a user logging into the application, When valid credentials and multi-factor authentication tokens are provided, Then the system grants access and encrypts all localized and in-transit health data using AES-256 standards.

3. PRODUCT BACKLOG PRIORITIZATION

Priority	User Story	Value	Dependency
1	US6.2 - Secure Authentication & Data Encryption	High	None
2	US1.1 - User Profile & BMI Onboarding	Critical	US6.2
3	US1.2 - Fitness Goal Setting	Critical	US1.1
4	US2.1 - Food & Macro Intake Logging	Critical	US1.1, Food Database API
5	US2.2 - Manual Workout Logging	High	US1.1
6	US5.1 - Real-Time Health Dashboard	High	US1.1, US2.1, US2.2
7	US4.1 - Wearable Device Pairing & Sync	High	Wearable SDK / API
8	US3.1 - AI Meal Plan Recommendation	Med-High	US1.1, US1.3, US2.1, ML Model
9	US3.2 - AI Workout Recommendation	Med-High	US1.2, US2.2, ML Model
10	US6.1 - Automated Reminders & Alerts	Medium	US2.1, US2.3
11	US2.3 - Water Intake Tracking	Medium	US1.1
12	US1.3 - Dietary Restrictions Config	Medium	US1.1
13	US4.2 - Sleep & Recovery Sync	Medium	US4.1
14	US3.3 - Dynamic Macro Adjustments	Medium	US3.1, US2.1
15	US5.2 - Weekly & Monthly Trend Reports	Low-Med	US5.1, Historical Data
16	US5.3 - Progress Report PDF Export	Low-Med	US5.2

Rationale: Security and baseline user onboarding (US6.2, US1.1, US1.2) are prioritized first, as downstream meal tracking and analytics depend strictly on accurate user biometric baselines. Food and activity tracking follow immediately to establish core tracking capability. AI recommendation algorithms and wearable integrations are implemented once adequate tracking data is captured, while detailed reporting and exports are scheduled in final iterations.

4. SPRINT BACKLOG PLANNING (SPRINT 1)

Sprint 1 targets the secure foundation, biometric profile creation, and core food/workout logging features, establishing the functional core necessary for all subsequent intelligent modules.

User Story	Development Tasks
US6.2	Security schema design → Auth API implementation (OAuth2/JWT) → Encryption service integration → Login/Register UI → Automated security testing
US1.1	User profile database model → Onboarding workflow UI → BMI/BMR calculation engine → Profile API endpoints → Unit testing
US1.2	Goal management schema → Goal selection UI → Caloric delta calculator integration → Backend integration tests
US2.1	Food database API client → Barcode scanner UI integration → Daily macro calculation engine → Local caching layer → Integration testing

5. STORY POINT ESTIMATION

Estimated via Planning Poker using the Fibonacci sequence (1, 2, 3, 5, 8, 13). Team disagreements were resolved through technical feasibility discussions regarding API integration and mobile UI complexity.

User Story	Points	Justification
US6.2	5	Standard authentication flow with AES-256 encryption overhead.
US1.1	3	Straightforward form input with standard biological math formulas.
US1.2	3	Extends onboarding with modest business logic for caloric deltas.
US2.1	8	High technical complexity: external API integration, barcode scanning, macro math.
US2.2	5	Exercise energy expenditure modeling and activity catalog UI.
US4.1	8	High complexity: Bluetooth LE hardware pairing and background sync protocols.
US3.1	13	Very high complexity: ML recommendation model development, training, and testing.
US5.1	5	UI dashboard layout and real-time data aggregation pipelines.

Sprint 1 Capacity: 19 points (US6.2 [5] + US1.1 [3] + US1.2 [3] + US2.1 [8]).

6. SPRINT GOAL & DELIVERABLES

Sprint Goal: "Establish a secure authentication foundation, complete biometric user onboarding with goal target settings, and deliver functional food macro logging, enabling users to register, configure personal health profiles, and log daily nutritional intake."

Deliverables:

- Secure OAuth2/JWT authentication system with encrypted data storage.
- Fully functional biometric onboarding flow calculating BMI, BMR, and caloric targets.
- Food logging interface integrated with barcode scanner and external nutritional database.
- Validated database models supporting user profiles, biometric history, and food logs.
- Demo-ready functional increment (Register → Profile Setup → Set Goals → Log Meal).

Definition of Done: All acceptance criteria satisfied and validated via automated unit/integration tests; code peer-reviewed and merged into staging; security compliance verified; functional demo approved by Product Owner during Sprint Review.

7. CONCLUSION

This workshop translates the Fitness and Diet Tracker problem statement into a structured, development-ready Agile plan. By defining the vision, mapping stakeholders, and decomposing requirements into testable Epics and User Stories, the plan keeps development tightly aligned with business value. The prioritized backlog, Sprint 1 plan, and story-point estimates give the team a clear, justified starting point and measurable delivery expectations, establishing a strong foundation for iterative development across future sprints.

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